

Standing-Wave Overlap Proxy for Alpha Preformation: P_α from Confined Flag Mode and Gaussian Cluster Trial

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Abstract

The companion nuclear-decay note [1] extracts $\log_{10} P_\alpha$ from Geiger–Nuttall intercepts and lifetime ratios on a 14-isotope set (^{212}Po – ^{244}Cm), without predicting preformation from interior structure. This sequel evaluates a *geometric/structural* standing-wave overlap proxy on the throat interval $[0, R]$: the $\ell = 0$ confined flag reduced radial mode $u_{\text{in}}(r) \propto r j_0(\pi r/R)$ against an alpha-cluster Gaussian/HO trial $u_\alpha(r) \propto r e^{-r^2/(2a_\alpha^2)}$ with $a_\alpha = r_0 \cdot 4^{1/3}$ and input $r_0 \approx 1.2$ fm. Normalized overlap P_{model} is compared to table-extracted $\log_{10} P_\alpha$. On the stated set we report $\langle \log_{10} P_{\text{model}} \rangle = -0.3905$, Pearson correlation $r = +0.8882$ versus extracted values, and RMS residual 1.7705 at $S = 1$. A single global spectroscopic factor $S \approx 2.70 \times 10^{-2}$ reduces RMS residual to 0.8220 without changing the correlation (affine shift). Absolute scale may require such a global S ; this is not a many-body uniqueness claim.

1 Motivation and scope

In [1], the Gamow factor W and transfer-matrix transmission fix the *slope* of alpha-decay systematics on the stated isotope set, while P_α remains an intercept-extracted nuclear-structure amplitude. The natural structural question is whether a confined standing-wave mode on $[0, R]$ overlaps an alpha-like radial packet enough to track the extracted $\log_{10} P_\alpha$ trend.

Remark (RC1 scope). P_{model} is a geometric/structural preformation *proxy*. It is not a first-principles shell-model or R-matrix spectroscopic factor, and absolute calibration may need one global factor S . Half-lives enter only through the extracted comparison column already published in [1]; they are not fitted into P_{model} .

2 Radial modes

2.1 Confined flag mode ($\ell = 0$)

Inside the throat the reduced radial amplitude for the first Dirichlet zero of j_0 is

$$u_{\text{in}}(r) = r j_0(kr), \quad k = \frac{\pi}{R}, \quad (1)$$

so $u_{\text{in}}(R) = 0$. Equivalently $u_{\text{in}}(r) = \sin(kr)/k$ on $(0, R]$.

2.2 Alpha-cluster trial

We take a Gaussian / harmonic-oscillator $0s$ packet truncated to the integration window $[0, R]$:

$$u_\alpha(r) = r e^{-r^2/(2a_\alpha^2)}, \quad a_\alpha = r_0 4^{1/3}, \quad (2)$$

with input $r_0 = 1.2$ fm, hence $a_\alpha \approx 1.905$ fm.

2.3 Overlap definition

$$P_{\text{model}} = \frac{\left| \int_0^R u_{\text{in}}(r) u_{\alpha}(r) dr \right|^2}{\left(\int_0^R u_{\text{in}}^2 dr \right) \left(\int_0^R u_{\alpha}^2 dr \right)}. \quad (3)$$

Optional global scale: $P_S = S P_{\text{model}}$, with S fitted so that $\langle \log_{10} P_S \rangle = \langle \log_{10} P_{\alpha}^{\text{ext}} \rangle$.

3 Numerical protocol

Integrals use Simpson quadrature on a uniform grid of $N = 4096$ points on $[0, R]$. Throat radii R and extracted $\log_{10} P_{\alpha}$ are taken from the 14-row table in [1]. Implementation path (repo): `code/palpa_overlap/palpa_c` results JSON: `docs/palpa_overlap_results.json`.

4 Results on the stated set

Table 1: **Standing-wave overlap vs extracted $\log_{10} P_{\alpha}$.** P_{model} from $\ell = 0$ overlap; S is one global factor.

Isotope	R (fm)	$\log_{10} P_{\text{model}}$	$\log_{10} P_{\alpha}^{\text{ext}}$	resid. (raw)	$\log_{10}(S P_{\text{model}})$	resid. (S)
^{212}Po	8.99	-0.3692	-1.53	+1.1608	-1.9373	-0.4073
^{214}Po	9.02	-0.3722	-1.19	+0.8178	-1.9403	-0.7503
^{216}Po	9.05	-0.3753	-1.04	+0.6647	-1.9434	-0.9034
^{218}Po	9.07	-0.3773	-0.82	+0.4427	-1.9454	-1.1254
^{220}Rn	9.10	-0.3804	-1.12	+0.7396	-1.9485	-0.8285
^{224}Ra	9.16	-0.3865	-1.54	+1.1535	-1.9546	-0.4146
^{226}Ra	9.18	-0.3885	-1.73	+1.3415	-1.9566	-0.2266
^{228}Th	9.21	-0.3916	-1.87	+1.4784	-1.9597	-0.0897
^{232}Th	9.26	-0.3967	-2.17	+1.7733	-1.9648	+0.2052
^{234}U	9.29	-0.3997	-2.64	+2.2403	-1.9678	+0.6722
^{238}U	9.34	-0.4048	-1.97	+1.5652	-1.9729	-0.0029
^{238}Pu	9.34	-0.4048	-3.28	+2.8752	-1.9729	+1.3071
^{240}Pu	9.36	-0.4068	-2.91	+2.5032	-1.9749	+0.9351
^{244}Cm	9.42	-0.4129	-3.61	+3.1971	-1.9810	+1.6290

4.1 Summary statistics

On the stated 14-isotope set ($r_0 = 1.2$ fm, $N = 4096$):

$$\begin{aligned} \langle \log_{10} P_{\text{model}} \rangle &= -0.3905 \quad (\sigma = 0.0142), \\ \langle \log_{10} P_{\alpha}^{\text{ext}} \rangle &= -1.9586 \quad (\sigma = 0.8656), \\ r(\log_{10} P_{\text{model}}, \log_{10} P_{\alpha}^{\text{ext}}) &= +0.8882, \\ \text{RMS}_{S=1} &= 1.7705, \quad S_{\star} = 2.703 \times 10^{-2}, \quad \text{RMS}_{S_{\star}} = 0.8220. \end{aligned}$$

Pearson r is unchanged by the global S (pure vertical shift in $\log_{10}$). S improves the absolute-scale RMS match by $\Delta\text{RMS} = +0.9485$.

Remark (Interpretation). Because a_{α} is fixed and R varies only weakly across the set, $\log_{10} P_{\text{model}}$ spans a narrow band ($\sigma \approx 0.014$) while extracted $\log_{10} P_{\alpha}$ spans $\sigma \approx 0.87$. The proxy therefore captures a mild R -driven geometric trend correlated with extracted preformation ($r \approx 0.89$), but leaves isotope-specific structure in the residual. A single global S absorbs mean offset; it does not remove residual structure.

5 Conclusion

We have computed the standing-wave overlap sequel requested in [1]: a confined $\ell = 0$ flag mode against a Gaussian alpha trial on $[0, R]$. On the stated set the proxy correlates with extracted $\log_{10} P_{\alpha}$ at $r = +0.8882$; absolute scale is improved by one global $S \approx 0.027$. This documents a geometric preformation trend, not a unique many-body prediction.

Remark (Next step). The flat $[0, R]$ measure and Gaussian trial are a geometric proxy. A sequel replaces them with the AdS-like throat weight $w(r) = R/r$ from the companion nuclear metric and a Woods–Saxon (optional Coulomb-tail) alpha trial [2]; baseline artifacts here are retained for comparison.

References

- [1] ACS Theoretical Physics Working Group, *Nuclear Decay as a Spectral Bifurcation of the Flag Condensate*, ACS Framework note (2026), `flag_condensate_nuclear_decay.tex` / `papers/notes/Flag_Condensate_Nuclear_Decay.tex`.
- [2] ACS Theoretical Physics Working Group, *Throat-Geometry Overlap for Alpha Preformation*, ACS Framework note (2026), `flag_condensate_palpha_throat_overlap.tex` / `papers/notes/Flag_Condensate_Palpha_Throat_Overlap.tex`.
- [3] G. Gamow, *Zur Quantentheorie des Atomkernes*, Z. Phys. **51**, 204 (1928).